

1: ① $\frac{2}{x}$ ② $\frac{x+y}{5}$ ③ $\frac{1}{2-a}$ ④ $\frac{x}{x-3}$

⑤ $\frac{x}{2x+1}$ ⑥ $\frac{x}{2} + y$ 属于分式的
有 _____

2: 当 $x = \underline{\hspace{2cm}}$ 时, 分式 $\frac{x^2-9}{x+3}$ 有意义

3: $\frac{1}{2x-3}$ 有意义的 x 的 _____

4: ① $\frac{x-1}{2x}$ 的值为 0, 求 $x = \underline{\hspace{2cm}}$

② $\frac{x^2-9}{x+3}$ 的值为 0, $x = \underline{\hspace{2cm}}$.

5: ① $\frac{9-x^2}{x^2-3x} = \underline{\hspace{2cm}}$

② $\frac{x^2-4x}{x^2-8x+16} = \underline{\hspace{2cm}}$

③ 已知 $-a+b-3=0$

求 $\frac{4(a-b)+8b}{a^2+2ab+b^2}$

⑤ $\frac{a^2-3a}{a^2+a} \div \frac{a-3}{a^2-1} \cdot \frac{a+1}{a+1}$

⑥ $\left(\frac{2xy}{-m^2}\right)^2 \div \frac{8x^2}{y^2} \cdot \left(\frac{-2m}{y^2}\right)^3$

⑦ $\frac{x^2+xy}{x-y} - \frac{y^2-3xy}{y-x}$

⑧ $\frac{4y}{(x+y)(x-y)} + \frac{5x}{x^2-y^2} + \frac{x}{y^2-x^2}$

④ $\frac{2a-4}{a^2-1} \div \frac{a-2}{a^2-2a+1}$

$$\textcircled{9} \frac{1}{2x^2y} - \frac{2}{3x^2} - \frac{3}{4xy^2}$$

$$\textcircled{14} \left(1 - \frac{1}{x+2}\right) \div \frac{x+1}{x^2+4x+4}.$$

$$\textcircled{10} \frac{x^2-1}{x^2+2x+1} + \frac{1}{x+1}$$

$$\textcircled{15} \left(\frac{m}{m-1} - 1\right) \div \frac{m^3-m}{m^2-2m+1}$$

$$\textcircled{11} \frac{3y}{2x+2y} + \frac{2xy}{x^2+xy}$$

$$\textcircled{16} \left(\frac{a+1}{2a-2} - \frac{1}{2a^2-2}\right) \div \frac{a^2}{a^2-1}.$$

$$\textcircled{12} \frac{x+2}{x^2-2x} - \frac{x-1}{x^2-4x+4}$$

$$\textcircled{17} 9(m+n)^2 - 3(m-n)(m+n).$$

$$\textcircled{18} (x^2+16y^2)^2 - 64x^2y^2$$

$$\textcircled{13} \frac{a^2}{a+1} - a + 1.$$

$$\textcircled{19} (y^2-1)^2 - 6(y^2-1) + 9.$$